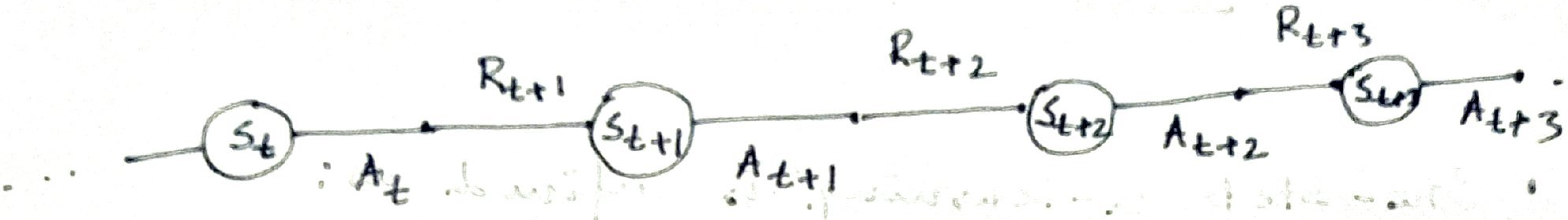
SARSA : On-policy TD Control



* For an on-policy method, we must estimate the state-value function $q_\pi(s, a)$ for current behavior of policy π and for all states s and action a.

* We consider transitions from state-action pair to state-action pair, rather than state to state.

* The value of state-action pairs are learned as follows:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t) \right].$$

This update is done after every transition from a nonterminal state S_t. If S_{t+1} is terminal, then $Q(S_{t+1}, A_{t+1})$ is defined as zero.

This rule uses every element of the quintuple of events:

$$\langle S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1} \rangle.$$

that make up a transition from one state-action pair to the next. This quintuple gives rise to the name SARSA for the algorithm.

SARSA algorithm

Initialize $Q(s,a)\ \forall s \in S,\ a \in A(s)$, arbitrarily and $Q(\text{terminal-state}, \cdot) = 0$.

Repeat (for each episode):

Initialize S

Choose A from S using policy derived from Q (e.g., ϵ-greedy).

Repeat (for each step of episode):

Take action A, observe R, S'

Choose A' from S' using policy derived from Q (e.g., ϵ-greedy).

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\right]$$

$S \leftarrow S'$; $A \leftarrow A'$; // $S \equiv S_t$; $S' \equiv S_{t+1}$;

Until S is terminal.

LAB: [Windy Gridworld]

Q-Learning: off-policy TD Control.

- One-step Q-learning is defined as:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\Big[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)\Big]$$

In this case, the learned action-value function Q, directly approximates q_*, the optimal action-value function, independent of the policy being followed.

Q-learning algorithm

Initialize $Q(s, a), \forall s \in \mathcal{S}, a \in \mathcal{A}(s)$, arbitrarily and $Q(\text{terminal-state}, \cdot) = 0$.

Repeat (for each episode):

  Initialize S

  Repeat (for each step of episode):

    Choose A from S using policy derived from Q (e.g., ϵ-greedy).

    Take action A, observe R, S'.

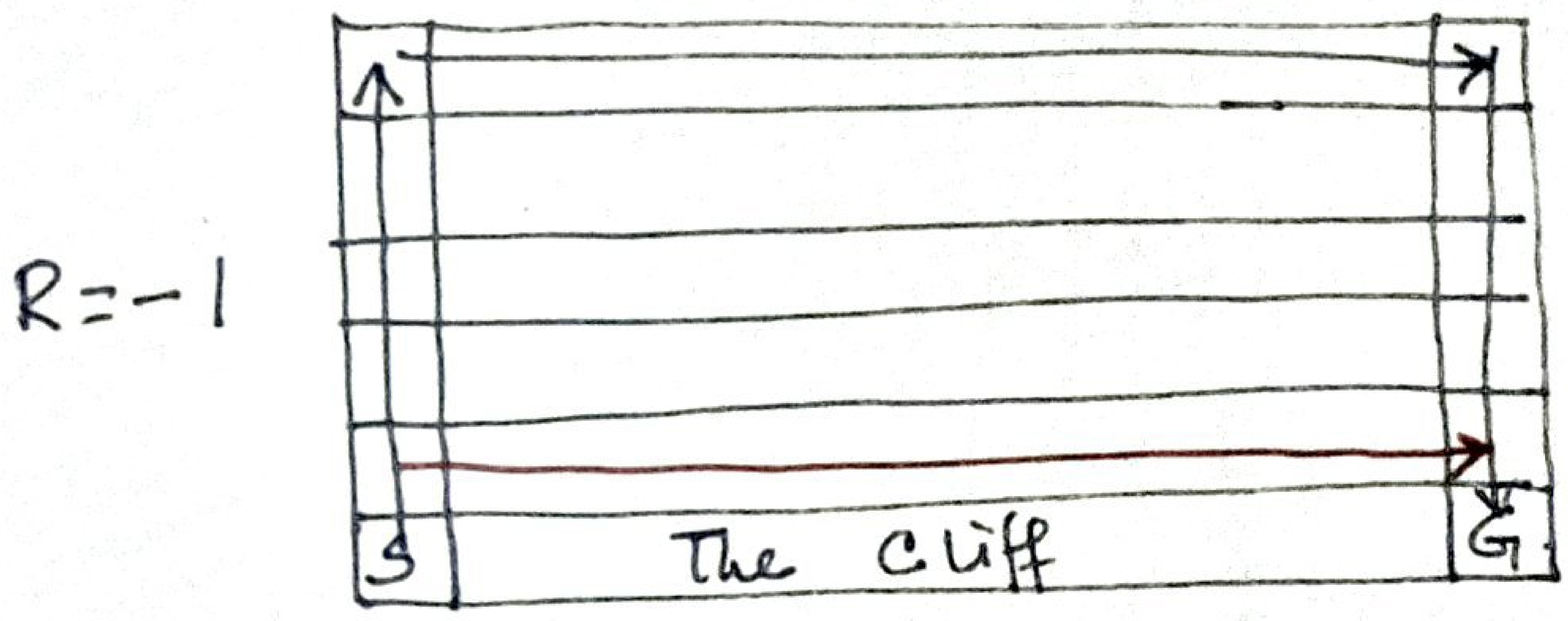
    $Q(S, A) \leftarrow Q(S, A) + \alpha\big[R + \gamma \max_a Q(S', a) - Q(S, a)\big]$.

    $S' \leftarrow S'$;

  Until S is terminal.

LAB: Cliff Walking.

• Cliff Walking:



Consider the grid world shown above. This is a standard undiscounted, episodic task, with start and goal states, and the usual actions causing movement up, down, right and left. Reward is -1 on all transitions except those into the the regiod marked "The Cliff". Stepping into this region incurs a reward of -100 and sends the agent instantly back to the start.

• Compare the performances between SARSA and Q-learning with ϵ-greedy action selection for $\epsilon = 0.1$, 0.01, 0.001. Show the plot on Reward per episode vs. No. of episode.

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